

R ABHISHEK

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Professional Summary

Research Engineer with 4 years of experience specializing in **Production-Grade Agentic AI** and **Multi-Agent Systems**. Architected and deployed high-traffic LLM solutions (NER, Summarization, Translation) serving 15k users, utilizing Model Distillation and vLLM to maintain sub-second latency. Expert in cost-efficient model alignment using ARPO/DPO and building end-to-end RAG pipelines featuring automated safety guardrails and performance benchmarking.

Technical Expertise

- **Generative AI:** LangGraph (Multi-Agent), AutoGen, vLLM (PagedAttention), DeepEval, RAGAS, Prompt Engineering.
- **Model Alignment & Tuning:** ARPO (Alignment via Reverse Preference Optimization), DPO, LoRA/QLoRA, Model Merging (MergeKit), Quantization (PTQ/QAT).
- **Machine Learning:** PyTorch, GNNs (Graph Neural Networks), Time-Series, Computer Vision, Reinforcement Learning.
- **Infrastructure:** AWS, Azure, Docker, Kubernetes, FastAPI, Vector DBs (Qdrant, Pinecone, Faiss, ChromaDB), CI/CD.

Professional Experience

Toshiba Research and Development

Bangalore, India

Senior Data Scientist

July 2023 – Present

- **Agentic Architecture & Orchestration:** Designed a multi-agent reasoning framework for automated Root Cause Analysis. Orchestrated agents with dynamic memory and tool-use, reducing manual investigative overhead by **70%** and saving **1.5x** in operational budget.
- **Cost-Optimized Inference:** Scaled internal LLM deployment using **vLLM** and **PagedAttention**; optimized GPU utilization for 32B+ parameter models, achieving a **3x throughput increase** and reducing token latency by 45%.
- **Production RAG & Evaluation:** Engineered a hybrid-search RAG engine for technical documentation; implemented **LLM-as-a-Judge** metrics (Faithfulness, Relevancy) and automated PII guardrails to meet enterprise compliance standards.
- **Advanced Reasoning Models:** Developed "Self-Correction" loops using **Qwen-32B** and **QWQ** style REACT and test-time compute strategies, improving task success rates in complex SQL and code-generation by 22%.

Data Scientist

July 2022 – July 2023

- **Custom Embedding Development:** Built a Japanese semantic embedding model using Graph-Walk algorithms to preserve hierarchy in POS data, outperforming OpenAI's text-embedding-3-small by **12%** in domain-specific tasks.
- **GNNs & Forecasting:** Implemented Temporal Attention Graphs for predictive maintenance in the power sector, achieving 94% accuracy in multi-node anomaly detection.

Samsung Research and Development Institute

Bangalore, India

Prism Researcher

July 2021 – Sept 2021

- **High-Performance Vision:** Conducted deep learning research for Free Space Detection in autonomous vehicles, securing funding for a secondary phase based on initial Signal-to-Noise (SNR) optimization benchmarks.

Engineering Projects

ARPO & MLPO: Advanced Tool-Augmented Alignment

- Developed an alignment suite using **ARPO** (Entropy-Based Adaptive Rollouts) to optimize tool-calling efficiency and **MLPO** for hierarchical leader-peer coordination. Applied step-level credit assignment to resolve reward sparsity in reasoning chains, achieving GPT-4 level synthesis in technical translation while reducing training compute by **50%** and VRAM by **30%**.

Enterprise Model Merging (MergeKit)

- Developed a composite model using **Spherical Linear Interpolation (SLERP)** to merge a specialized coding model with a general reasoning model, resulting in a dual-capability agent without the cost of separate deployments.

Real-Time Fault Detection SaaS

- Built a full-stack dashboard (React/FastAPI/MongoDB) for energy forecasting, utilizing **Active Learning** to reduce the need for labeled data by 40%.

Publications & Awards

- **Gold Medalist (M.Tech):** Graduated 1st in cohort at IIITDM Kurnool.
- **Global Rank 8th:** IEEE ICETCI Competition (Hosted by **ISRO**) for satellite data ML extraction.
- **IEEE Publication:** "Active learning based semantic segmentation." Published at 42nd IEEE IGARSS, Malaysia (2022).

Education

Indian Institute of Information Technology Kurnool | M.Tech, Data Analytics and Decision Sciences

Gold Medalist

CMR School of Engineering | B.Tech, Computer Science and Engineering

First Class